Day 118: The Week 17 Build — Model Selection and Cross-Validation Strategies

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 17, Day 7 | Complete study guide

Watch: Nested Cross Validation - Algorithm Explained (Machine Learning Mastery) | Time Series Cross Validation Explained (The NextGen Ai Frontier)

Script: `day118_model_selection.py` · every number in this guide is copied by program from `day118_run_output.txt`, the real output of that script.

The week's six exercise parts, copied in verbatim, become the models, and three ways of cutting folds are tested on real data. Stratified folds halve the wobble of a CV estimate on a rare class (0.0778 to 0.0445). Fewer folds underestimate (2-fold misses by -0.0727, leave-one-out by -0.0093). On time series, shuffled folds report an error about six times smaller than kNN really makes on the future CO2 record (0.521 against 7.232), and forward-chaining is closer (2.932) but still short. Two results did not follow the usual story, and the guide says so: the best CV score was not optimistic here, and forward-chaining did not choose better than shuffled folds.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

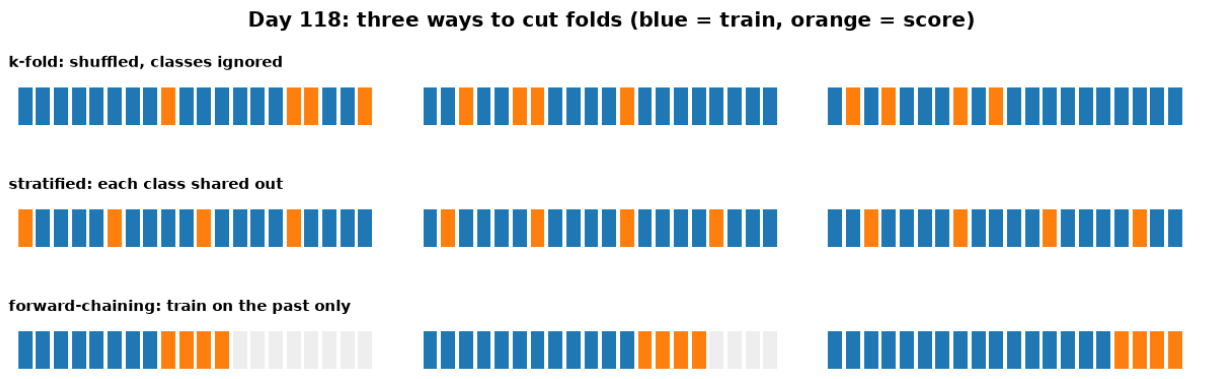
Cross-validation (CV) cuts the training rows into k **folds**, trains on k−1 of them, scores on the one left out, and averages. How the folds are cut matters: **k-fold** shuffles and ignores the labels; **stratified** folds give every fold its share of each class; **forward-chaining** trains only on the past and scores on the next block of time.

Nested CV repeats the whole choice of model inside each outer fold, so the outer score judges the *procedure*, choice included. Day 118 is the week's **build**: the exercise parts of Days 112 to 117 are copied in unchanged and used as the models.

REAL-WORLD SCENARIO

A weather service tests its forecaster by hiding random past days and predicting them from the days around them. The scores look superb, because the model saw tomorrow when predicting today. Tested only on days after its training data, it scores worse, and that worse score is the one that matters.

SYSTEM DIAGRAM



Three ways to cut five folds; three of the five rounds are shown for each.

Data: digits (a 9 against the rest), breast cancer (whole, and with malignant tumours subsampled to 5%), and two real time series bundled with the statsmodels library: weekly CO2 at Mauna Loa (59 missing weeks filled

by straight lines) and yearly sunspot counts. statsmodels supplies only the data.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. Checks: the parts are verbatim

DEFINITION — start here, no prior knowledge assumed

The script parses every exercise file of the week and checks that each function and class appears in itself, character for character. Day 115 and Day 117 each carried a copy of Day 113's sqdist; one copy is enough.

REAL-WORLD SCENARIO

Checking that the parts on the assembly line are exactly the parts that passed inspection.

IMPLEMENTATION

```
me = open(os.path.abspath(__file__), encoding="utf-8").read(); found, total, dup = 0, 0, []
for d in range(112, 118):
    src = open(os.path.join(HERE, f"day{d}_exercise.py"), encoding="utf-8").read(); lines =
        src.split("\n")
    for node in ast.parse(src).body:
        if isinstance(node, (ast.FunctionDef, ast.ClassDef)) and node.name not in HELPERS:
            text = "\n".join(lines[node.lineno - 1:node.end_lineno])
            if (d, node.name) in SKIP:
                dup.append(f"{node.name} (day{d})"); continue
            total += 1; found += text in me
```

ACTUAL OUTPUT

```
functions and classes from day112_exercise.py to day117_exercise.py found here, character for
    character: 24 of 24
left out as duplicates of Day 113's identical sqdist: sqdist (day115), sqdist (day117); the
    hand-check helpers of Day 116 were not copied
Actual output from this lesson's run.
```

What it means

All 24 of 24 parts are present unchanged: Day 112's SMO solver, Day 113's kernels, Day 114's Naive Bayes, Day 115's kNN and PCA, Day 116's stacking and Day 117's metrics and SMOTE. Today the SVM, kNN and Day 117's metrics do the work.

WHY CHOOSE THIS?

The build cannot drift from the parts it claims to use.

WHY NOT THIS? (TRADEOFFS)

A verbatim check proves the copy, not that the parts suit today's problems.

2. k-fold against stratified folds on a rare class

DEFINITION — start here, no prior knowledge assumed

The same CV, reshuffled 50 times. With plain k-fold, the number of rare rows in a fold is left to chance; stratified folds deal each class round the folds like cards, so every fold gets its share.

REAL-WORLD SCENARIO

Five teams picked at random from a class with three left-handers: some teams get none. Dealing the left-handers out first fixes that.

IMPLEMENTATION

```
def kfold(n, k, seed):
    return np.array_split(np.random.RandomState(seed).permutation(n), k)
def stratified(y, k, seed):
    rng = np.random.RandomState(seed); folds = [[] for _ in range(k)]
    for c in np.unique(y):
        idx = rng.permutation(np.where(y == c)[0])
        for i, j in enumerate(idx):
            folds[i % k].append(j)
    return [np.array(sorted(f)) for f in folds]
```

ACTUAL OUTPUT

```
Day 117's metrics on kNN (k = 5) predictions; the spread is how much the CV estimate changes when only the
    shuffle changes
digits: 9 against the rest: 1797 rows, 180 rare (0.100)
  k-fold     : rare rows per fold 24 to 50; folds with none 0 of 250; CV F1 0.9527, spread across reshuffles
      (std) 0.0048
  stratified : rare rows per fold 36 to 36; folds with none 0 of 250; CV F1 0.9520, spread across reshuffles
      (std) 0.0043
breast cancer, 5% malignant: 376 rows, 19 rare (0.051)
  k-fold     : rare rows per fold 0 to 9; folds with none 4 of 250; CV F1 0.7207, spread across reshuffles
      (std) 0.0778
  stratified : rare rows per fold 3 to 4; folds with none 0 of 250; CV F1 0.7264, spread across reshuffles
      (std) 0.0445
Actual output from this lesson's run.
```

What it means

On breast cancer with 19 malignant rows, plain k-fold gave folds with 0 to 9 rare rows, and 4 of 250 folds had none at all, so recall could not even be measured there. Stratified folds always hold 3 to 4, and the CV F1 wobbles about half as much between reshuffles (0.0445 against 0.0778). On digits, with 180 rare rows, chance already spreads them well enough and the two are level (0.0048 and 0.0043). Stratifying costs nothing, and it matters most when the rare class is small.

WHY CHOOSE THIS?

A steadier estimate, and no fold left without the class you care about.

WHY NOT THIS? (TRADEOFFS)

Classification only; regression and time need other cuts.

3. How good is a CV estimate?

DEFINITION — start here, no prior knowledge assumed

Forty training sets of 150 digit images each. Each set's CV estimate is compared with the same model's accuracy on all the other 1,647 images, which is as close to the truth as this data allows.

REAL-WORLD SCENARIO

Judging a cook from a few dishes: tasting more dishes gives a fairer picture, but only of what this cook made this week.

MATHEMATICS

$$bias = mean(CV\ estimate - true\ accuracy); \quad error\ std = spread\ of\ (CV\ estimate - true\ accuracy)$$

ACTUAL OUTPUT

```
each training set's CV estimate is compared with the model's accuracy on all the other 1,647
    images: the truth, as near as it can be measured
         folds  mean estimate  mean truth     bias  error std
       2-fold         0.8528      0.9255  -0.0727     0.0376
       5-fold         0.9007      0.9255  -0.0248     0.0257
      10-fold         0.9093      0.9255  -0.0162     0.0258
 leave-one-out         0.9162      0.9255  -0.0093     0.0239
fewer folds train on fewer rows, so they underestimate; every estimate still misses a given
    training set's true accuracy by a few points
Actual output from this lesson's run.
```

What it means

Fewer folds mean smaller training parts, and kNN does worse with less data, so 2-fold underestimates by -0.0727 on average while leave-one-out is off by only -0.0093. The spread is the humbling part: for any single training set, even leave-one-out misses the truth by a typical 0.0239, and 5-fold and 10-fold are level (0.0257, 0.0258). On 150 rows, a single CV score is good to a few points either way, and it leans low.

WHY CHOOSE THIS?

Five or ten folds give most of leave-one-out's accuracy at a fraction of the cost.

WHY NOT THIS? (TRADEOFFS)

One model on one dataset; the size of the bias depends on how fast the model improves with data.

4. The cost of choosing: best CV score, nested CV and the test

DEFINITION — start here, no prior knowledge assumed

Thirteen candidates (nine RBF SVMs and four kNNs) are scored by 5-fold CV on the training part, and the best is chosen. Is the winner's CV score a fair estimate of its test accuracy? Nested CV runs the whole choice again inside each of five outer folds.

REAL-WORLD SCENARIO

Thirteen students sit a practice exam, and the top score is announced as the class's future grade. Luck helped whoever came top.

SPECIFICATION

Splits	10 stratified 70/30 splits of breast cancer (569 rows); the test part is never seen by any choice.
Candidates	RBF SVM with gamma 0.003, 0.03, 0.3 and C 0.1, 1, 10 (Days 112-113); kNN with k = 1, 5, 15, 35 (Day 115).
Interval	A bootstrap over the 10 splits (Day 99); the splits share rows, so it is too narrow.

IMPLEMENTATION

```
outer = stratified(yt, 5, 100 + s); nested = []
for i in range(5):
    va = outer[i]; ft = np.concatenate([outer[j] for j in range(5) if j != i])
    inner = {n: cv_score(f, Xt[ft], yt[ft], stratified(yt[ft], 5, s)) for n, f in C.items()}
    nested.append(np.mean(C[max(inner, key=inner.get)](Xt[ft], yt[ft])(Xt[va]) == yt[va]))
```

ACTUAL OUTPUT

```
13 candidates: 9 RBF SVMs (gamma x C) and 4 kNNs; 5-fold stratified CV on the training part
    chooses one
best CV score (what is often reported):  0.9726
nested CV estimate:                       0.9678
the chosen model's test accuracy:         0.9784
  best CV    minus test: -0.0057 [-0.0129, +0.0023]
  nested CV  minus test: -0.0105 [-0.0176, -0.0028]
models chosen: [('SVM gamma 0.003 C 10', 4), ('SVM gamma 0.03 C 1', 1), ('SVM gamma 0.03 C 10',
    5)]
Actual output from this lesson's run.
```

What it means

The textbook warning is that the best of many CV scores is optimistic. It did not show here: the best CV score (0.9726) sat *below* the chosen model's test accuracy (0.9784), by -0.0057, with an interval that includes zero. Nested CV was lower still (-0.0105). A likely reason, not tested here: every CV model trains on only 80% of the training part, which costs accuracy, and with only 13 candidates the luck of picking the top one may be small. Only three SVMs were ever chosen. Nested CV's job is to remove the optimism of choosing; when there is little optimism to remove, it simply adds the cost of training on less data.

WHY CHOOSE THIS?

Nested CV is the honest estimate of a whole procedure, choice included.

WHY NOT THIS? (TRADEOFFS)

Here 5 x 13 x 5 = 325 fits per split, and it was not needed.

5. Time: shuffled folds against forward-chaining

DEFINITION — start here, no prior knowledge assumed

Predict the next value of a series from its last 8. The first 80% of each series is for choosing and estimating; the last 20% is held back as the real future. Shuffled 5-fold mixes past and future rows; forward-chaining trains only on what came before each block.

REAL-WORLD SCENARIO

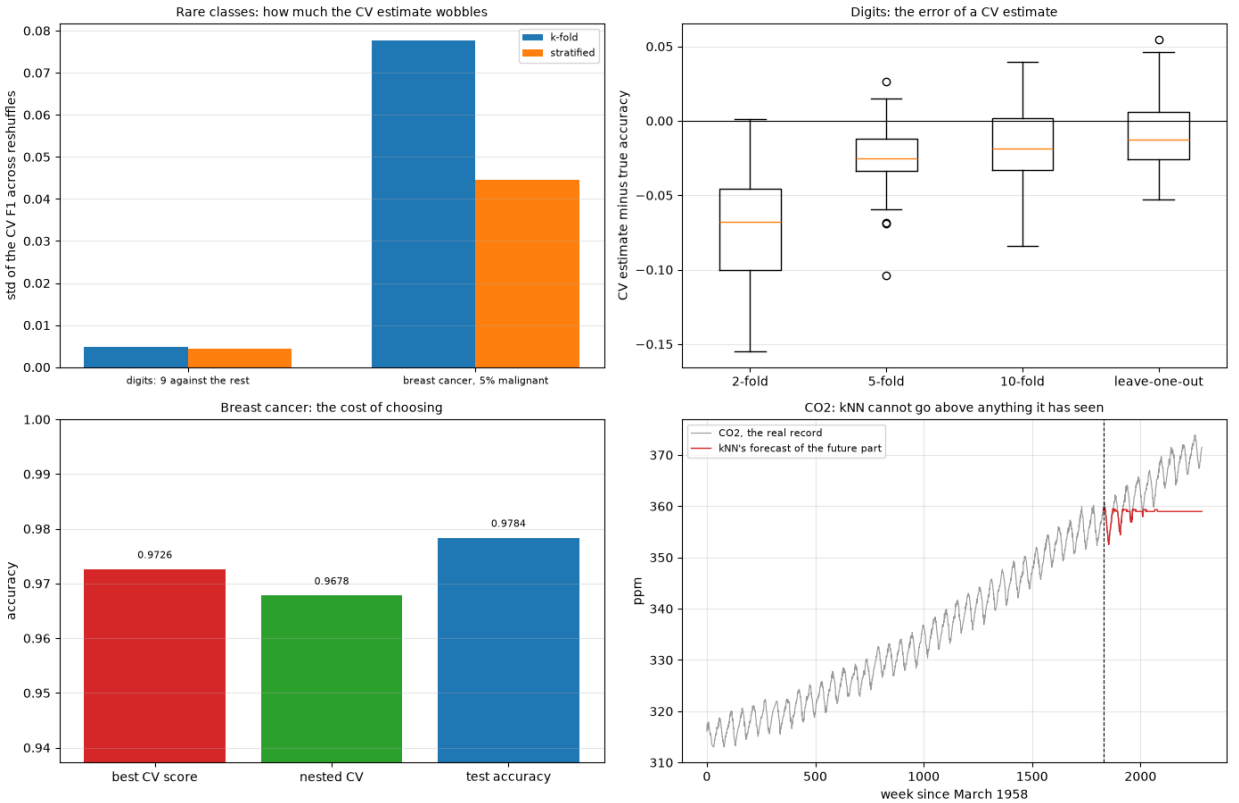
Predicting tomorrow's temperature is easy if you are allowed to look at the day after tomorrow.

IMPLEMENTATION

```
def forward(n, k, min_train):
    edges = np.linspace(min_train, n, k + 1).astype(int)
    return [(np.arange(edges[i]), np.arange(edges[i], edges[i + 1])) for i in range(k)]
def lagged(series, lags):
    X = np.stack([series[i:len(series) - lags + i] for i in range(lags)], 1)
    return X, series[lags:]
def knn_reg(X, y, k):
    return lambda Z: y[np.argpartition(sqdist(Z, X), k - 1, axis=1)[:, :k]].mean(1)
```

ACTUAL OUTPUT

```
predict the next value from the last 8; the first 80% of each series is for choosing and
    estimating, the last 20% is the real future
forward-chaining: 5 blocks, each predicted by a model trained only on everything before it
CO2 (weekly, ppm): 2284 values; past 1820 rows, future 456 rows; RMSE in the series' own units
  model              shuffled 5-fold  forward-chaining  the real future
  kNN, k = 5                   0.521             2.932            7.232
  linear on lags               0.438             0.457            0.466
Sunspots (yearly): 309 values; past 240 rows, future 61 rows; RMSE in the series' own units
  model              shuffled 5-fold  forward-chaining  the real future
  kNN, k = 5                  16.677            17.641           29.108
  linear on lags              14.627            14.596           20.320
Actual output from this lesson's run.
```



Top left: how much a rare-class CV estimate wobbles. Top right: the error of a CV estimate on digits. Bottom left: the cost of choosing. Bottom right: kNN's forecast of the CO2 future part.

What it means

For kNN on CO2, shuffled folds report an RMSE of 0.521 ppm; the real future gives 7.232. Every shuffled test row has neighbours from the weeks on either side of it, so kNN can copy them. In the future part, CO2 rises above anything in the training record, and kNN can only average values it has seen (bottom right of the figure: the forecast is flat). Forward-chaining is more honest (2.932) but still short, probably because the future also lies further from the training data than any of its blocks. The linear model on lags can extrapolate, and all three of its numbers agree (0.438, 0.457, 0.466). On sunspots every estimate is too low against the real future (29.108 and 20.320); the last 61 years of that record were harder to predict than the years before.

WHY CHOOSE THIS?

Forward-chaining is the only fold scheme that respects time.

WHY NOT THIS? (TRADEOFFS)

It trains on less data in its early blocks, and it still assumes the future behaves like the past.

6. Choosing a model for the future

DEFINITION — start here, no prior knowledge assumed

kNN's k, chosen from six values by shuffled CV, by forward-chaining, and, in hindsight, by the real future.

REAL-WORLD SCENARIO

Picking a forecaster by how well they 'predicted' last year with last year's answers in hand.

ACTUAL OUTPUT

```
CO2 (weekly, ppm):
              kNN k 1    kNN k 3    kNN k 5   kNN k 10   kNN k 20   kNN k 40
  shuffled        0.677      0.554      0.521      0.523      0.568      0.709
  forward         3.081      2.957      2.932      3.022      3.254      3.756
  future          6.944      7.305      7.232      7.374      7.626      7.858
  shuffled CV chooses kNN k 5 (future RMSE 7.232); forward-chaining chooses kNN k 5 (7.232); best in hindsight
      kNN k 1 (6.944)
Sunspots (yearly):
              kNN k 1    kNN k 3    kNN k 5   kNN k 10   kNN k 20   kNN k 40
  shuffled       19.184     16.504     16.677     16.623     17.602     20.537
  forward        22.718     18.570     17.641     17.457     19.755     23.878
  future         31.091     26.706     29.108     31.250     35.103     38.848
  shuffled CV chooses kNN k 3 (future RMSE 26.706); forward-chaining chooses kNN k 10 (31.250); best in
      hindsight kNN k 3 (26.706)
Actual output from this lesson's run.
```

What it means

Forward-chaining fixes the *level* of the estimate but did not fix the *choice*. On CO2 both strategies choose k = 5 (future RMSE 7.232), while k = 1 was best in hindsight (6.944). On sunspots shuffled CV happened to choose the best k (26.706), and forward-chaining chose k = 10 (31.250). With five blocks, and a future unlike the past, neither scheme ranks the candidates reliably; what forward-chaining buys is a warning that the error will be large.

WHY CHOOSE THIS?

Shows that an honest estimate and a good choice are different goals.

WHY NOT THIS? (TRADEOFFS)

Two series and one family of models; with a model that extrapolates, the picture can change.

Interview preparation — technical questions

Q: When do you use stratified folds?

A: On classification, always, and especially with a rare class: here they halved the wobble (0.0778 to 0.0445).

Q: How many folds?

A: 5 or 10: 2-fold underestimated by 0.0727 on digits; 5- and 10-fold were within 0.025 and about as steady as leave-one-out.

Q: How do you cross-validate a time series?

A: Forward-chaining: train on the past, score on the next block. Shuffled folds gave 0.521 against a real 7.232.

Q: What is nested CV for?

A: Estimating a whole procedure, model choice included; the inner loop chooses, the outer loop scores.

Q: Why must the test set stay out of every choice?

A: Any choice made with it turns it into training data, and its score stops being an estimate.

Interview preparation — theoretical questions

Q: Why does fewer folds mean a pessimistic estimate?

A: Each model trains on fewer rows: with 2 folds, on half of them.

Q: Why is the best CV score usually optimistic?

A: The maximum of noisy scores is biased upward; here it was not, perhaps because the CV models train on less data.

Q: Why does shuffling leak on time series?

A: Neighbouring time points are near-copies, so the model sees the answer's neighbours, including later ones.

Q: Can kNN forecast a trend?

A: No: it averages values it has seen, so a rising series leaves its forecast behind (7.232 ppm on CO2).

Key takeaway and what's next

> **KEY TAKEAWAY**
> How the folds are cut decides what a CV score means. Stratify on rare classes, use five or ten folds, keep the test part out of every choice, and on time series train only on the past. Forward-chaining made the estimate honest; it did not make the choice better, and the best CV score was not optimistic here: both are reminders to measure rather than assume.

- Rare class: stratified folds 0.0445 wobble against 0.0778; no empty folds.
- Digits: 2-fold bias -0.0727, leave-one-out -0.0093.
- Breast cancer: best CV 0.9726, nested 0.9678, test 0.9784.
- CO2, kNN: shuffled 0.521, forward 2.932, real future 7.232 ppm.

Suggested watch order, if starting from zero

- 1. Nested Cross Validation - Algorithm Explained (Machine Learning Mastery) — the inner and outer loops of nested CV.
- 2. Time Series Cross Validation Explained (The NextGen Ai Frontier) — why shuffled folds fail on time series.

Day 119 preview

- **Week 18 begins: k-means from scratch**, with a real elbow-method demonstration.